

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, customer segments, and subscription behavior to guide strategic decisions.



Dataset at a Glance

3,900

Total Purchases

Transactional records analyzed

18

Features

Demographics, purchases, behavior

37

Missing Values

In Review Rating column only

50

Locations

Geographic coverage

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season, Size, Color), and shopping behavior (Discount, Previous Purchases, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

01

Load & Explore

Imported dataset via `pandas`; used `df.info()` and `.describe()` for structure and summary statistics.

02

Clean Missing Data

Imputed missing **Review Rating** values using median rating per product category.

03

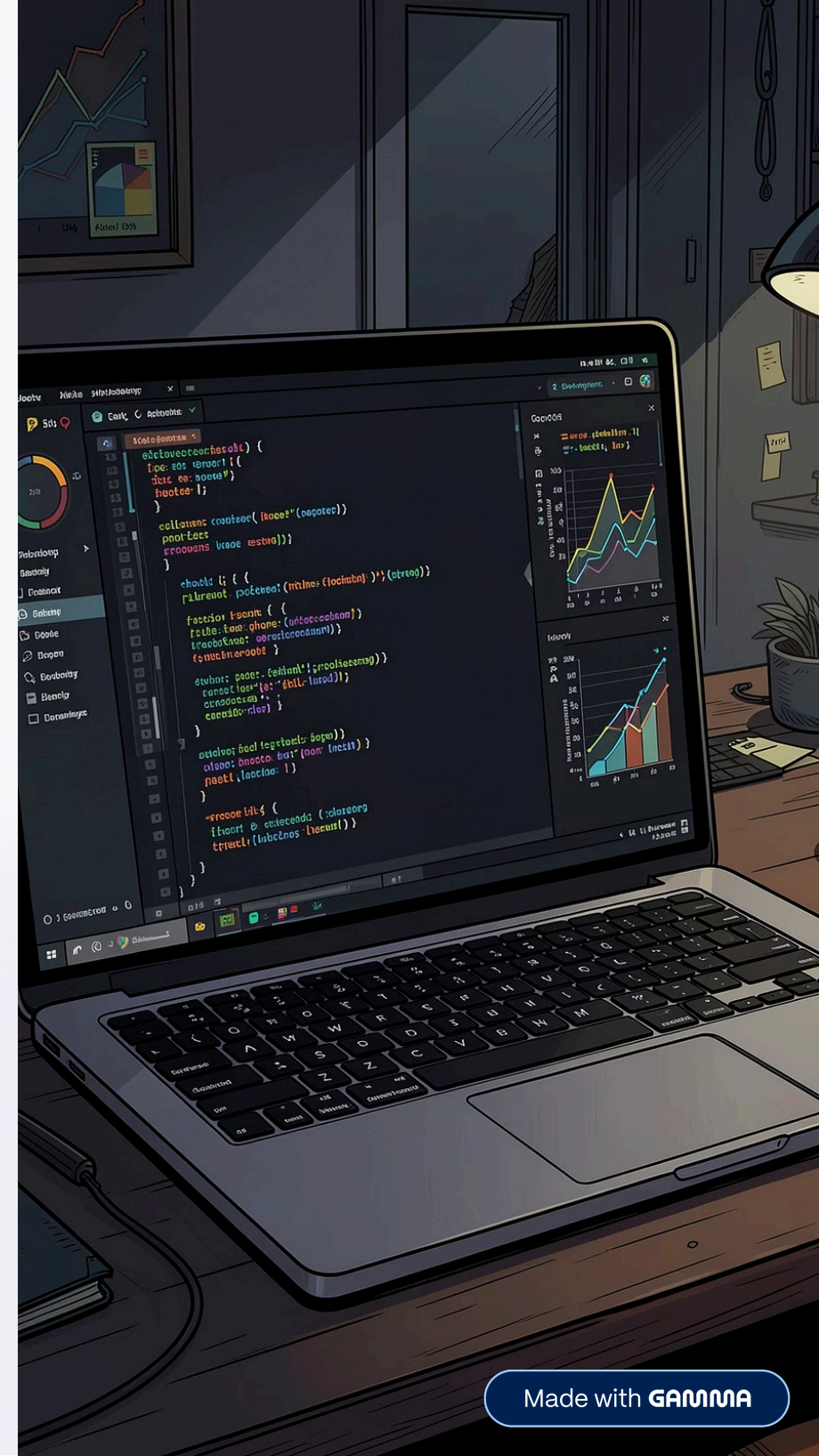
Feature Engineering

Created `age_group` by binning ages and `purchase_frequency_days` from purchase data. Dropped redundant `promo_code_used` column.

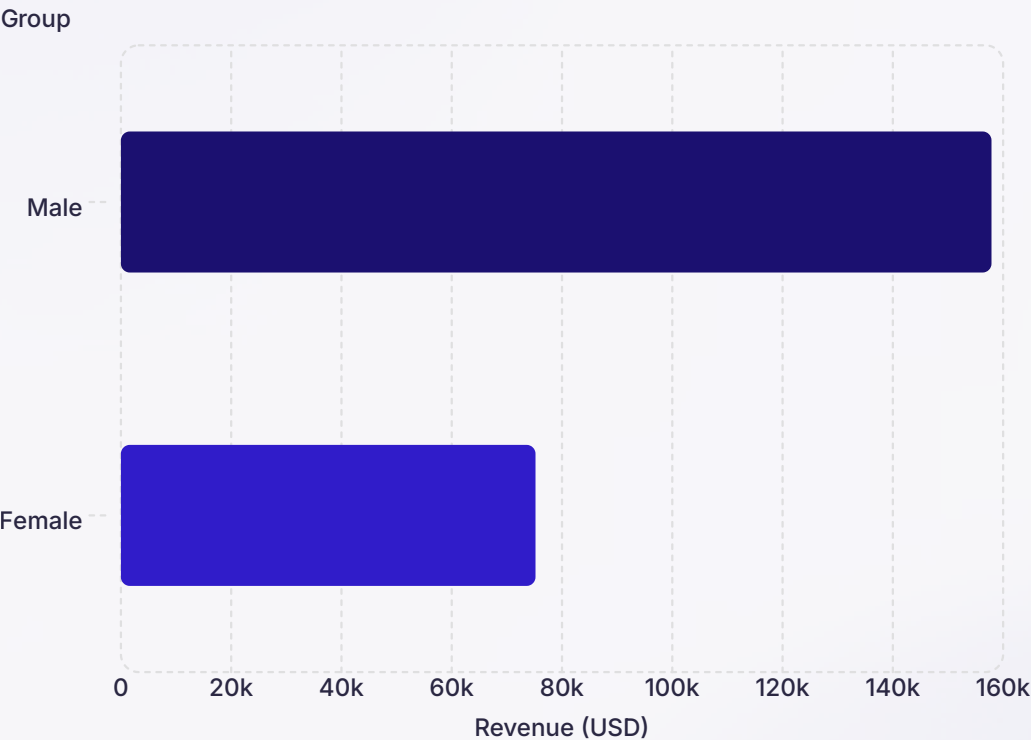
04

Database Integration

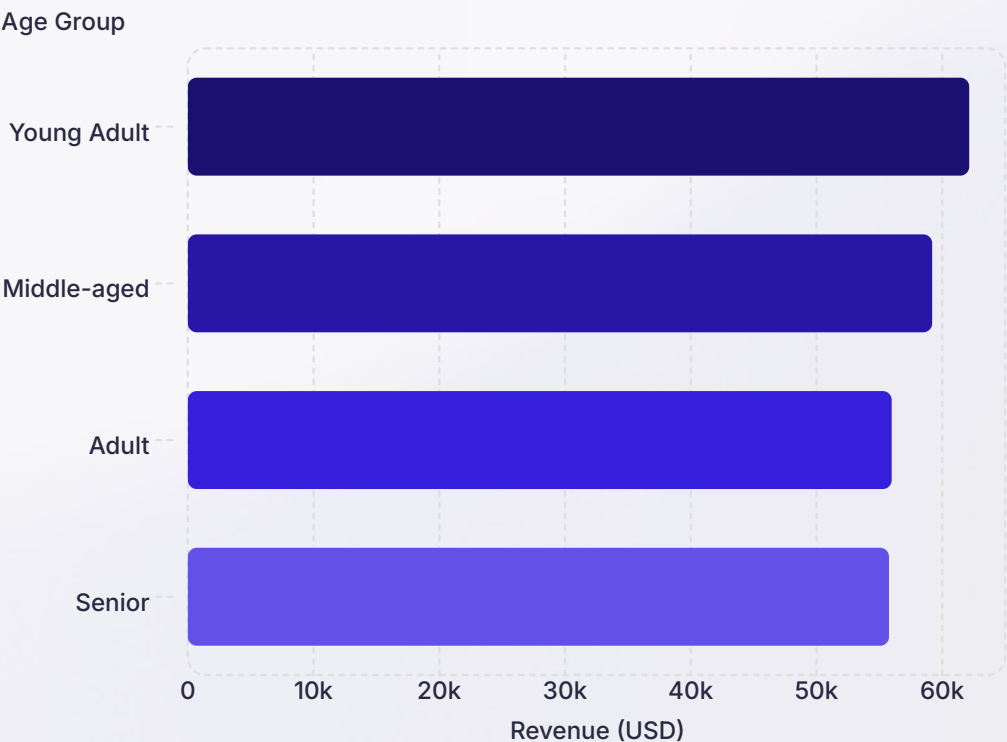
Loaded cleaned DataFrame into **PostgreSQL** for SQL-based analysis.



Revenue by Gender & Age Group



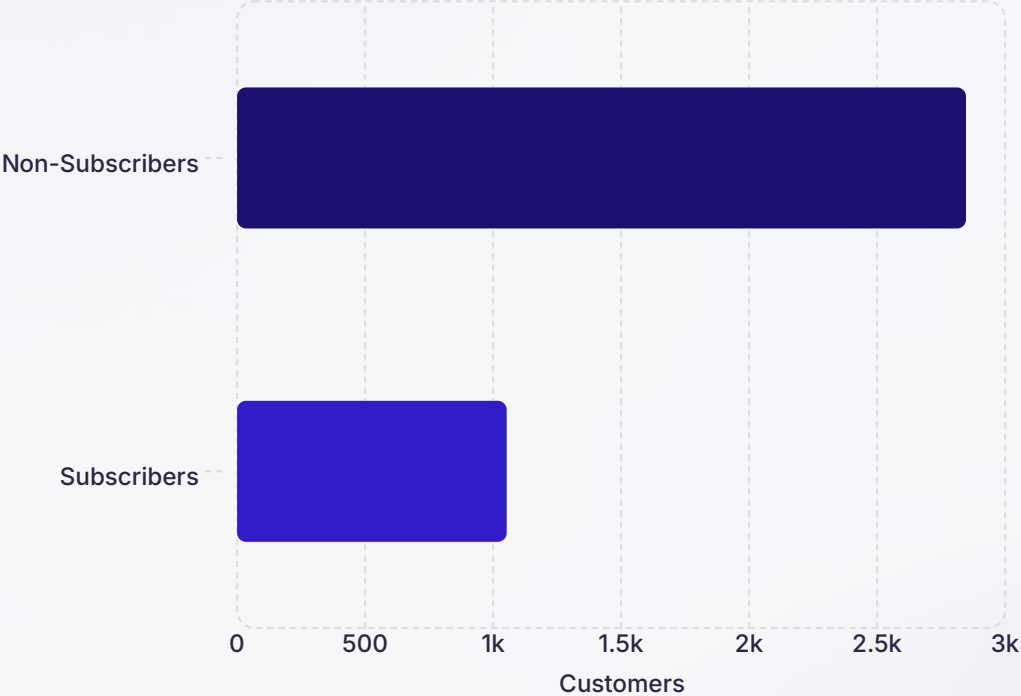
Revenue by Age Group



Male customers generated **2.1x more revenue** than female customers (\$157,890 vs \$75,191). Young Adults led all age groups in revenue contribution at \$62,143, though revenue is relatively evenly distributed across age segments.

Subscription & Customer Segmentation

Subscription Status



Key Findings

73% Non-Subscribers

2,847 non-subscribers generated \$170,436 in total revenue vs. \$62,645 from 1,053 subscribers.

Average Spend Nearly Equal

Subscribers: \$59.49 avg spend; Non-subscribers: \$59.87 — minimal difference.

Repeat Buyers

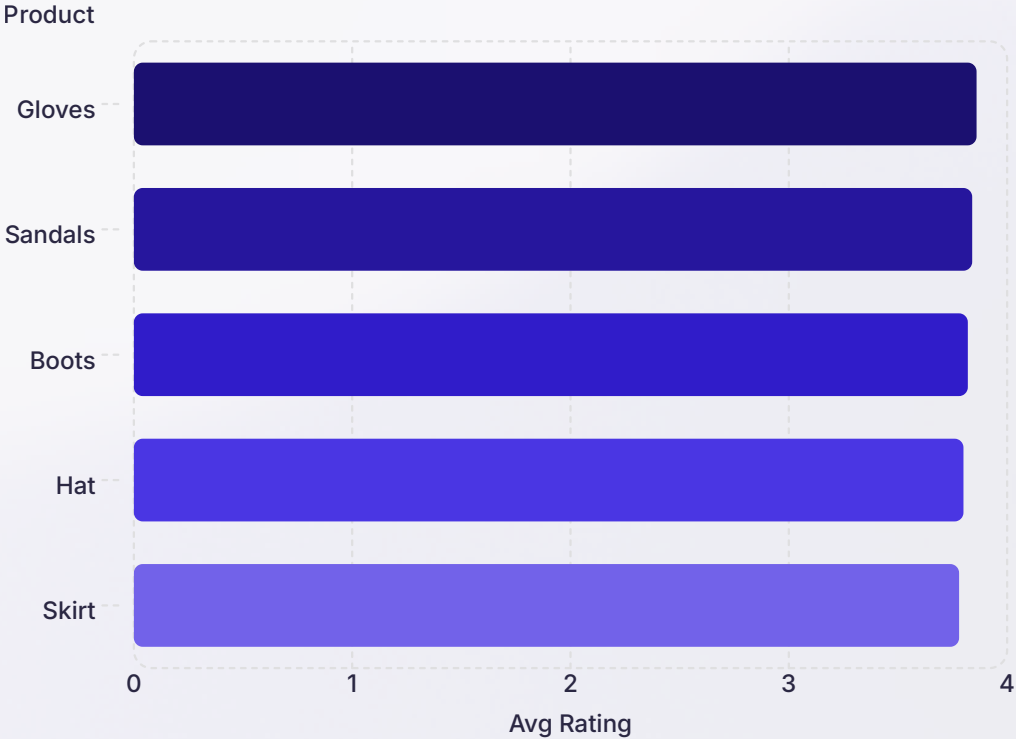
Of customers with 5+ purchases, 958 subscribe vs. 2,518 who don't.

Top Products by Category & Rating

Top 3 Products per Category

Category	Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163

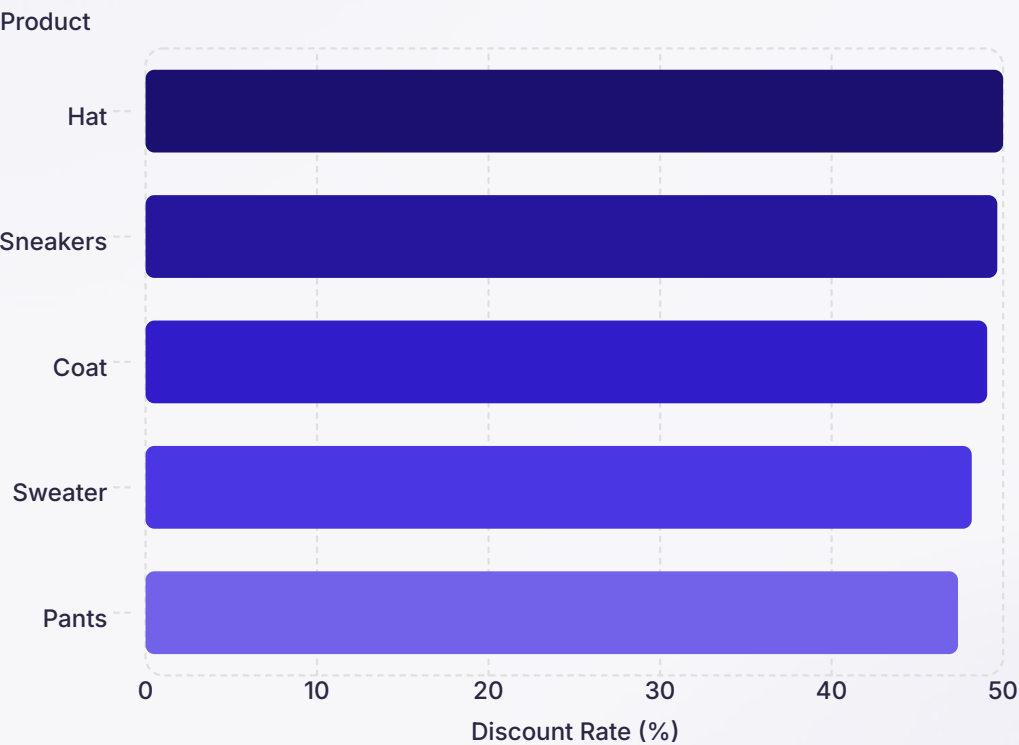
Top 5 by Review Rating



Overall average review rating: **3.75**

Discount Behavior & Shipping

Most Discount-Dependent Products



Shipping Type Comparison

Express shippers spend slightly more on average. **839 customers** used discounts yet still spent above the overall average purchase amount.

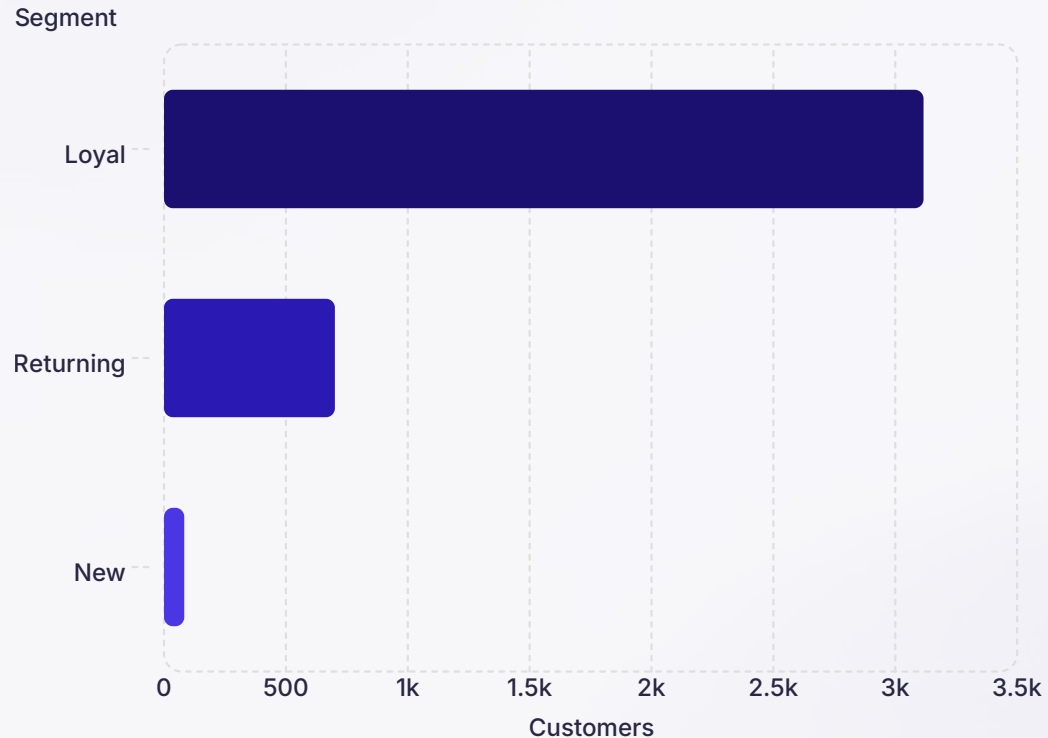
Standard Shipping

Avg purchase: \$58.46

Express Shipping

Avg purchase: \$60.48

Customer Segmentation



Segment Breakdown

→ **Loyal — 3,116 customers**

The dominant segment, representing the vast majority of the customer base.

→ **Returning — 701 customers**

Active repeat buyers with growth potential toward loyalty.

→ **New — 83 customers**

Smallest segment; key target for onboarding and retention campaigns.



Power BI Dashboard

3.9K Customers

Total customer base tracked

\$59.76 Avg Purchase

Mean transaction value

3.75 Avg Rating

Overall review score

The interactive dashboard visualizes **subscription status** (27% Yes, 73% No), **revenue by category** (Clothing ~\$90K, Accessories ~\$60K, Footwear ~\$30K, Outerwear ~\$15K), and **revenue by age group** (Young Adult ~\$55K leading). Filters include Gender, Category, and Shipping Type for drill-down analysis.



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Customer Loyalty Programs

Reward repeat buyers to advance them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control — Hats and Sneakers show ~50% discount dependency.



Targeted Marketing

Focus campaigns on high-revenue age groups (Young Adults) and express-shipping users.